
How Far Can Robots Trust Action Chunks? Horizon Adaptation via Action-Expert Evidence

Anonymous Author(s)

Affiliation

Address

email

Abstract

Executing fixed-length action chunks is standard in robot foundation policies, but the execution horizon—how many predicted actions to commit before the next replan—is a latent variable, not a static hyper-parameter. Existing adaptive chunking methods reduce horizon selection to a scalar decision driven by a single heuristic cue, misaligning the problem with the action expert whose internal generation dynamics and boundary continuity ultimately determine prefix reliability. We formulate horizon selection as an action-expert posterior over executable prefixes, and propose a training-free *Action-aware Horizon Selector* (AHS) that scores candidate horizons using two complementary evidence sources derived from the policy’s own predictions: intra-chunk spectral stability (Fourier analysis of the denoising velocity trace) and inter-chunk continuity (speed uniformity at the boundary between executed history and predicted prefix). An online Beta posterior with kernel forgetting tracks phase-dependent horizon preferences. We further introduce a lightweight *Query-based Horizon Adapter* (QHA) that distills AHS preferences into a learned posterior, enabling an amortized horizon prior that can be combined with fresh online evidence. Across RoboTwin2.0 and RoboCasa GR1 Tabletop, AHS improves every simulated base policy, with the largest AHS-only gains of +6.6 percentage points on $\pi_{0.5}$ and +9.7 percentage points on Qwen3GR00T; fusing QHA with AHS raises the RoboTwin2.0 gain on $\pi_{0.5}$ to +9.5 percentage points. On four real-world household tasks, $\pi_{0.5}$ + AHS improves the equal-task mean normalized process score from 50.4% to 57.5% (+7.1 percentage points). Ablations confirm that both evidence terms, the temporal posterior, and QHA contribute meaningfully.

1 Introduction

Vision-Language-Action (VLA) [1–6] and World-Action-Model (WAM) [7–9] policies generate actions in fixed-length chunks to amortize inference cost and promote temporal coherence. However, open-loop execution of the entire predicted chunk implicitly treats all future actions as equally reliable until the next observation, an assumption that often fails in real-world manipulation [10–12]. Early actions may be stable and executable, while later actions accumulate compounding prediction errors, become invalid under new observations, or lose contact alignment as the task phase shifts. This tension creates a *trust boundary* problem: how many predicted actions can we reliably execute before the next replan? As Fig. 1 (top) shows in a real bread handover-manipulation rollout, the answer is phase-dependent—free-space approach tolerates longer commitment, while delicate transport and contact demand earlier replanning.

Since the reliable prefix length is unobserved and phase-dependent, we view the execution horizon—the number of predicted actions committed before the next policy call—as a latent variable to infer online rather than a fixed hyperparameter. Fig. 1 (bottom) supports this view empirically: the

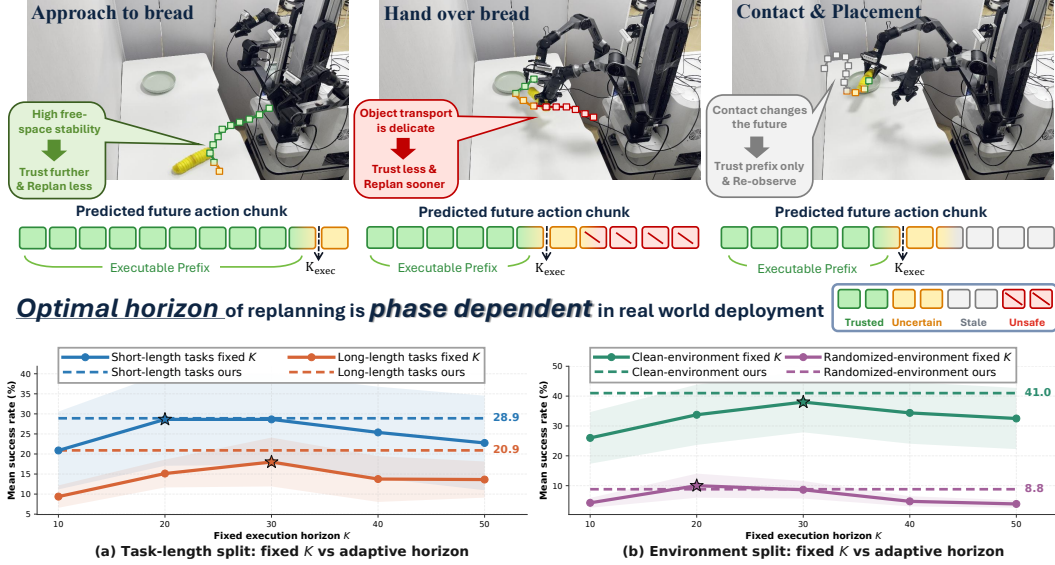


Figure 1: **Fixed execution horizons are brittle across tasks and settings.** **Top:** the reliable executable prefix is phase dependent in a real bread-manipulation rollout. **Bottom:** fixed horizons $K \in \{10, 20, 30, 40, 50\}$ are compared with AHS+QHA on π_0 RoboTwin2.0 evaluations, split by (a) task length and (b) environment setting.

best fixed horizon K varies substantially across task lengths and evaluation settings. Recent work has begun to adapt the execution horizon at inference time, using signals such as attention limits, action entropy, or cross-horizon consensus. However, these methods largely treat horizon selection as a one-shot scalar readout: a proxy is computed at the current replan step and directly mapped to an execution length. This view leaves three aspects under-modeled. First, it does not maintain a belief over multiple candidate prefixes, and therefore cannot represent uncertainty over how far the current chunk remains reliable. Second, the proxy is often only indirectly related to prefix executability: a low-entropy or high-consensus chunk may still contain unstable action-generation dynamics or introduce a discontinuity when stitched after the executed history. Third, independent per-step decisions provide little memory of phase-dependent horizon preferences, despite manipulation naturally alternating between stable free-space motion and contact-sensitive phases.

We instead formulate horizon adaptation as *posterior inference over executable prefixes*. A candidate prefix is executable when it is both *internally stable* under the action expert’s generation dynamics and *externally compatible* with the recently executed action history. Under this view, action-derived cues are not used as deterministic rules for selecting K ; they serve as likelihood evidence for updating a belief over candidate horizons. Maintaining such a posterior lets the robot express uncertainty, accumulate evidence across replans, and shift its horizon preference as the task moves between free-space, contact, transport, and placement phases.

This formulation leads to two complementary instantiations. First, we propose a training-free *Action-aware Horizon Selector* (AHS), which performs online posterior inference without modifying the base policy. At each replan, AHS evaluates candidate prefixes using two evidence sources derived from the policy’s own predictions: *intra-chunk spectral stability*, which detects high-frequency instability in the denoising velocity trace, and *inter-chunk continuity*, which measures whether the predicted prefix can be smoothly stitched to the executed action history. These evidence terms define an action-expert likelihood over executable prefixes, which is accumulated by a lightweight Beta posterior with kernel forgetting to track non-stationary phase-dependent horizon preferences.

Second, we introduce a *Query-based Horizon Adapter* (QHA) as an amortized prior for the same posterior inference problem. While AHS is deployment-ready and training-free, it must score a sparse candidate set online and infer horizon preferences reactively from the current rollout. QHA distills AHS-derived posterior preferences into a learned dense horizon distribution conditioned on multimodal context and action-latent tokens. At deployment, this learned prior is not used to replace

69 AHS; instead, it is combined with fresh online action-expert evidence, allowing the robot to exploit
70 familiar horizon patterns while remaining responsive to scene-specific deviations.

71 We evaluate AHS in simulation and real-world deployment. In simulation, we test on RoboTwin2.0
72 and RoboCasa GR1 Tabletop across base policies spanning several flow-based action experts. AHS
73 alone improves every simulated base policy, with the largest AHS-only gains of +6.6 percentage
74 points on $\pi_{0.5}$ in RoboTwin2.0 and +9.7 percentage points on Qwen3GR00T in RoboCasa. When
75 QHA is fused with AHS, the RoboTwin2.0 gain reaches +9.5 percentage points on $\pi_{0.5}$. We
76 further deploy the system on an AgileX COBOT Magic ALOHA-style platform over four household
77 manipulation tasks, where $\pi_{0.5}$ + AHS improves the equal-task mean normalized process score from
78 50.4% to 57.5%, an absolute gain of +7.1 percentage points. Ablations show that the two evidence
79 terms capture complementary failure modes, the online posterior supports phase-aware adaptation,
80 and AHS+QHA further improves AHS when QHA is used as an amortized prior.

81 In summary, our contributions are threefold. (1) We formulate adaptive chunk execution as posterior
82 inference over executable prefixes, reframing horizon selection from a deterministic chunk-size rule
83 into an action-expert reliability estimation problem. (2) We instantiate this formulation with two
84 complementary mechanisms: AHS, a training-free online selector that uses intra-chunk spectral
85 stability and inter-chunk continuity as likelihood evidence for a phase-aware Beta posterior; and
86 QHA, a lightweight amortized prior that distills AHS-derived preferences into a dense horizon
87 distribution and combines with fresh online evidence at deployment. (3) We validate the framework
88 across RoboTwin2.0, RoboCasa GR1 Tabletop, and real-world deployment, with representative gains
89 of +9.5 percentage points on $\pi_{0.5}$ in RoboTwin2.0, +9.7 percentage points on Qwen3GR00T in
90 RoboCasa, and +7.1 percentage points in real-world normalized process score; ablations confirm
91 that the two evidence terms, temporal posterior, and amortized prior each contribute to performance.

92 2 Horizon-Aware Evidence from the Action Expert

93 2.1 Preliminaries

94 Under a common mainstream architecture, flow-based VLA/WAM policies pair a VLM backbone
95 with an action expert [1, 2, 7], while WAMs additionally include a video-prediction head [8, 9]. They
96 amortize control by predicting an action chunk and executing only a prefix before replanning [10, 13].
97 Given visual observation o_t , proprioceptive state s_t , and instruction ℓ , the VLM backbone produces a
98 multimodal context $c_t = f_\theta(o_t, s_t, \ell)$ for the action expert. At replan step t , the base policy outputs

$$\hat{\mathbf{A}}_t = [\hat{\mathbf{a}}_{t,1}, \dots, \hat{\mathbf{a}}_{t,H}] \in \mathbb{R}^{H \times d_a}. \quad (1)$$

99 Rather than executing the full chunk, the controller chooses an execution horizon $K_t \in \mathcal{K} \subseteq$
100 $\{1, \dots, H\}$ and executes only the prefix $\hat{\mathbf{a}}_{t,1:K_t}$ before acquiring the next observation. We call a
101 prefix *executable* when it is internally stable under the action expert’s generation dynamics and
102 externally compatible with the recently executed action history. Thus, reliability is evaluated from
103 action-expert evidence, not VLM confidence alone.

104 At each replan, the policy can be called once with trace recording, yielding both the predicted chunk
105 and generation trace, $(\hat{\mathbf{A}}_t, \mathcal{F}_t) = \pi_\theta(o_t, s_t, \ell)$. For flow-based action experts [14], this trace contains
106 the velocity or denoising states [15] used for horizon scoring while the base action generator remains
107 unchanged. We write the trace and executed history as

$$\mathcal{F}_t = \{\mathbf{v}_{t,\tau} \in \mathbb{R}^{H \times d_a}\}_{\tau=0}^{T-1}, \quad \mathcal{H}_t = [\mathbf{a}_{t-N_t}^{\text{exec}}, \dots, \mathbf{a}_{t-1}^{\text{exec}}], \quad (2)$$

108 where τ indexes the internal sampling step and N_t is the number of valid history steps. This section
109 defines two evidence terms for each $k \in \mathcal{K}$: intra-chunk stability from $\mathcal{F}_t$ and inter-chunk continuity
110 from $(\mathcal{H}_t, \hat{\mathbf{A}}_t)$, which Sec. 3 fuses into an online posterior over executable horizons.

111 2.2 Intra-Chunk Spectral Stability

112 The first evidence term measures whether the action expert generates a candidate prefix through a
113 smooth and stable velocity trajectory along the action-horizon axis. For a candidate horizon $k \in \mathcal{K}$,
114 we compare velocity-prefix spectra across candidate horizons and apply a one-dimensional Fourier
115 transform along the action-horizon dimension for each action dimension $j \in \{1, \dots, d_a\}$ and discrete

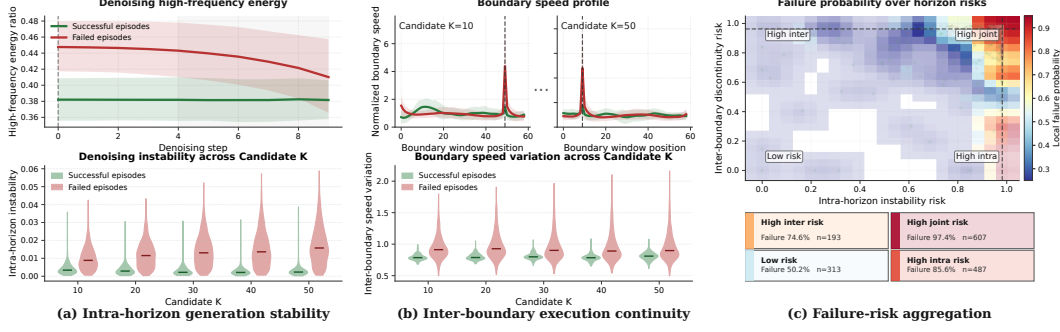


Figure 2: **Metric rationality of horizon-aware evidence.** Episode-level π_0 analysis on RoboTwin2.0 with fixed $H = 50$. (a) Intra-chunk spectral stability. (b) Inter-chunk boundary continuity. (c) Joint intra/inter risk and failure probability.

frequency $\omega \in \Omega$:

$$\hat{\mathbf{V}}_{t,\tau}^{(k)}(\omega, j) = \text{FFT}_h\left(\bar{\mathbf{v}}_{t,\tau}^{(k)}[:, j]\right)_\omega, \quad \text{where} \quad \bar{\mathbf{v}}_{t,\tau}^{(k)} = \text{Pad}_H(\mathbf{v}_{t,\tau,1:k,:}) \in \mathbb{R}^{H \times d_a}, \quad (3)$$

Let $\Omega_{\text{hi}} \subset \Omega$ denote frequencies above a cutoff fraction c_{cut} of the discrete frequency grid. The high-frequency ratio is the fraction of spectral energy, aggregated over action dimensions, that lies in this high-frequency band:

$$r_{\text{hi},t,\tau}(k) = \frac{\sum_{\omega \in \Omega_{\text{hi}}} \sum_{j=1}^{d_a} \left| \hat{\mathbf{V}}_{t,\tau}^{(k)}(\omega, j) \right|^2}{\sum_{\omega \in \Omega} \sum_{j=1}^{d_a} \left| \hat{\mathbf{V}}_{t,\tau}^{(k)}(\omega, j) \right|^2}. \quad (4)$$

A large high-frequency ratio indicates that the candidate prefix contains rapidly varying velocity components along the future-action dimension, which is a proxy for local oscillation or unstable generation inside the chunk.

The trace provides not only a final action sequence but also a temporal evolution of this spectrum during generation. We therefore score a candidate by how much its high-frequency ratio deviates from an early-generation baseline. With a short window W_τ over the first denoising steps, let $\bar{r}_{\text{hi},t,0}(k) = \frac{1}{W_\tau} \sum_{\tau=0}^{W_\tau-1} r_{\text{hi},t,\tau}(k)$. The raw intra-chunk instability is the root mean square (RMS) deviation of the later high-frequency ratio from this early-generation baseline:

$$z_{\text{intra},t}(k) = \left[\frac{1}{T - W_\tau} \sum_{\tau=W_\tau}^{T-1} (r_{\text{hi},t,\tau}(k) - \bar{r}_{\text{hi},t,0}(k))^2 \right]^{1/2}. \quad (5)$$

Because the scale of $z_{\text{intra},t}$ varies across tasks, policies, and action normalizations, we min-max normalize it within $\mathcal{K}$ and set $q_{\text{intra},t}(k) = \exp[-\alpha_{\text{intra}} z_{\text{intra},t}(k)]$, where $\alpha_{\text{intra}} > 0$ controls the penalty strength. Thus $q_{\text{intra},t}(k)$ is close to one for prefixes whose generation trace remains spectrally stable and decreases as the candidate prefix exhibits unstable high-frequency dynamics. Fig. 2 (a) validates this evidence term empirically: failed episodes exhibit elevated high-frequency energy (**top**) and larger RMS deviation from the early-generation baseline across candidate horizons (**bottom**).

2.3 Inter-Chunk Continuity

To prevent velocity discontinuities at the transition boundary, the second evidence term measures whether a candidate prefix can be smoothly spliced after the actions that have already been executed. A locally stable chunk can still be undesirable if its first few actions create a sharp discontinuity at the replan boundary. Let W_h be the continuity window. For each candidate k , we concatenate the suffix of the executed history with the current future prefix:

$$\mathbf{S}_t^{(k)} = [\text{Suffix}_{(W_h-k)_+}(\mathcal{H}_t), \hat{\mathbf{a}}_{t,1}, \dots, \hat{\mathbf{a}}_{t,k}]. \quad (6)$$

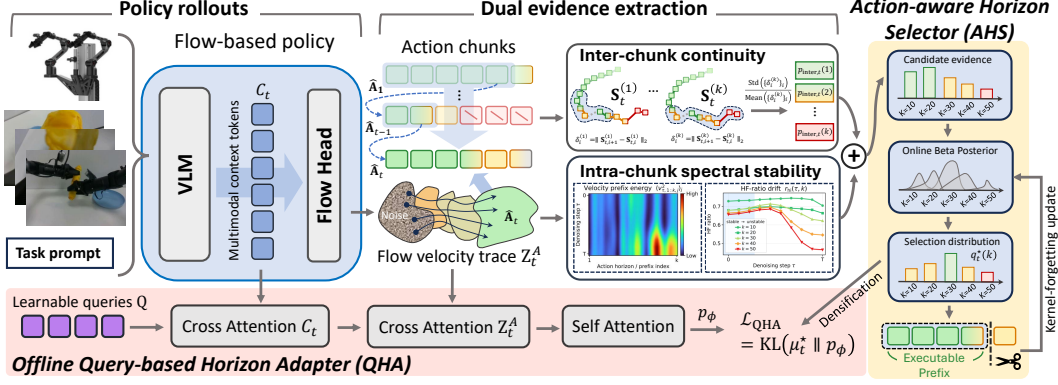


Figure 3: **Overview of horizon adaptation.** AHS uses action-expert evidence to update an online posterior over executable horizons, while QHA distills AHS preferences into an amortized prior.

Here $(x)_+ = \max(0, x)$. We use a speed-uniformity proxy: let $\delta_i^{(k)} = \|\mathbf{S}_{t,i+1}^{(k)} - \mathbf{S}_{t,i}^{(k)}\|_2$ be the first-difference speed along the stitched trajectory. We denote the resulting raw inter-chunk discontinuity by $u_{\text{inter},t}(k)$:

$$u_{\text{inter},t}(k) = \text{Std}(\{\delta_i^{(k)}\}_i) / \text{Mean}(\{\delta_i^{(k)}\}_i). \quad (7)$$

This statistic penalizes irregular speed profiles after stitching the history and the candidate future. A smaller value means that the candidate prefix is more compatible with the executed trajectory, while a larger value indicates a likely boundary jump, stop-and-go behavior, or over-aggressive correction.

As with the intra-chunk evidence term, we min-max normalize the raw score within $\mathcal{K}$ and set $q_{\text{inter},t}(k) = \exp[-\beta_{\text{inter}} \tilde{u}_{\text{inter},t}(k)]$, where $\beta_{\text{inter}} > 0$ controls the penalty strength. Fig. 2(b) confirms this empirically: failed episodes show larger speed spikes at the boundary (**top**) and higher speed variation across all candidate horizons (**bottom**).

2.4 Dual-Evidence Fusion

The two evidence terms capture complementary failure modes. The intra-chunk score tends to reject prefixes whose future actions are internally oscillatory or underspecified. The inter-chunk score rejects prefixes that are individually plausible but incompatible with the recently executed motion. We combine them into a candidate quality score

$$q_{\text{mix},t}(k) = (1 - \lambda)q_{\text{intra},t}(k) + \lambda q_{\text{inter},t}(k), \quad \lambda \in [0, 1]. \quad (8)$$

In our default configuration, we set $\lambda = 0.5$, while $\lambda = 0$ and $\lambda = 1$ recover intra-only and inter-only ablation settings. The fused score should not be interpreted as a deterministic chunk-size rule. Rather, it is an action-expert likelihood over candidate executable prefixes.

For the episode-level analysis in Fig. 2(c), we convert the two quality scores into risks, $1 - q_{\text{intra}}$ and $1 - q_{\text{inter}}$, and summarize each episode by the 75th percentile over its replans. The median-split quadrants show a clear monotonic pattern: the low-risk region has a 50.2% failure rate, high intra-only and high inter-only regions rise to 85.6% and 74.6%, and the high-joint-risk region reaches 97.4%. This supports our use of the two evidence terms as complementary evidence for dynamic horizon selection.

3 Action-aware Horizon Selection and Query-based Adaptation

Sec. 2 defines action-expert evidence for a single replan. We now use this evidence in two complementary ways: AHS maintains a training-free online posterior over candidate horizons, while QHA distills AHS-derived preferences into a lightweight amortized prior without modifying the base action generator. In both cases, K_t denotes the number of future actions executed before the next policy call.

Table 1: **Per-task success rates on RoboTwin2.0.** Easy/Hard denote clean/randomized settings. Yellow columns use AHS and blue columns use AHS+QHA. Parentheses report percentage-point deltas over Base.

Task	π_0 [1]		π_0 (AHS)		π_0 (AHS+QHA)		$\pi_{0.5}$ [2]		$\pi_{0.5}$ (AHS)		$\pi_{0.5}$ (AHS+QHA)		Fast-WAM [7]		Fast-WAM (AHS)		
	Easy	Hard	Easy	Hard	Easy	Hard	Easy	Hard	Easy	Hard	Easy	Hard	Easy	Hard	Easy	Hard	
Blocks Ranking RGB	19	0	28 (+9)	1 (+1)	31 (+12)	5 (+5)	36	20	48 (+12)	29 (+9)	56 (+20)	33 (+13)	99	98	100 (+1)	100 (+2)	
Handover Block	41	10	41 (+0)	5 (-5)	64 (+23)	9 (-1)	44	14	44 (+0)	14 (+0)	32 (-12)	10 (-4)	94	81	91 (-3)	82 (+1)	
Handover Mic	100	2	100 (+0)	24 (+22)	100 (+0)	30 (+28)	98	64	100 (+2)	56 (-8)	98 (+0)	59 (-5)	100	99	100 (+0)	100 (+1)	
Hanging Mug	17	3	14 (-3)	9 (+6)	23 (+6)	10 (+7)	14	8	13 (-1)	12 (+4)	13 (-1)	14 (+6)	66	64	71 (+5)	69 (+5)	
Place A2B Left	24	1	34 (+10)	0 (-1)	39 (+15)	1 (+0)	41	4	47 (+6)	4 (+0)	47 (+6)	15 (+11)	95	95	93 (-2)	93 (-2)	
Place Bread Basket	11	8	18 (+7)	16 (+8)	15 (+4)	10 (+2)	30	17	50 (+20)	29 (+12)	55 (+25)	41 (+24)	91	91	93 (+2)	94 (+3)	
Place Bread Skillet	15	2	23 (+8)	5 (+3)	23 (+8)	2 (+0)	24	10	36 (+12)	19 (+9)	44 (+20)	19 (+9)	91	91	94 (+3)	94 (+3)	
Place Can Basket	33	5	22 (-11)	2 (-3)	33 (+0)	3 (-2)	35	15	50 (+15)	29 (+14)	55.2 (+20.2)	34 (+19)	71	63	69 (-2)	67 (+4)	
Average	32.5	3.9	35.0 (+2.5)	7.8 (+3.9)	41.0 (+8.5)	8.8 (+4.9)	40.2	19.0	48.5 (+8.3)	24.0 (+5.0)	50.0 (+9.8)	28.1 (+9.1)	88.4	85.2	88.9 (+0.5)	87.4 (+2.1)	
Overall		18.2		21.4 (+3.2)		24.9 (+6.7)		29.6		36.2 (+6.6)		39.1 (+9.5)		86.8		88.1 (+1.3)	

3.1 Action-aware Horizon Selector

Candidate scoring. For every $k \in \mathcal{K}$, we compute $q_{\text{intra},t}(k)$ from the intra-chunk spectral score (Sec. 2.2) and $q_{\text{inter},t}(k)$ from the inter-chunk continuity score (Sec. 2.3), then fuse them with Eq. (8). The resulting score $q_{\text{mix},t}(k) \in [0, 1]$ is a soft estimate of the current reliability of executing k actions from the candidate chunk.

Online reliability posterior. A single replan score can be noisy because the deployed robot may pass through non-stationary phases such as approach, contact, transport, and placement. We therefore maintain an online Beta posterior over the reliability of each candidate, following the Thompson sampling framework [16]: for every $k \in \mathcal{K}$, AHS stores the Beta state $\xi_t(k) \sim \text{Beta}(a_t(k), b_t(k))$, initialized by $a_0(k) = b_0(k) = 1$. This posterior is not a supervised task-success model. It is a lightweight memory of recent horizon reliability under the current episode dynamics. At replan step t , we draw a Thompson sample and combine it with the instantaneous action-expert quality:

$$\hat{\xi}_t(k) \sim \text{Beta}(a_t(k), b_t(k)), \quad s_t(k) = \hat{\xi}_t(k) q_{\text{mix},t}(k). \quad (9)$$

The non-negative scores are converted into a temperature-scaled selection distribution, and expected-round selection chooses the execution length:

$$\mu_t(k) \propto s_t(k)^{1/T_{\text{sel}}}, \quad \sum_{k \in \mathcal{K}} \mu_t(k) = 1, \quad K_t = \text{clip}_{[1, H_t^{\text{avail}}]} \text{round} \left[\sum_{k \in \mathcal{K}} k \mu_t(k) \right]. \quad (10)$$

If all scores degenerate numerically, μ_t falls back to the uniform distribution over valid candidates.

Execution rule. Expected-round selection smooths horizon changes across sparse candidates. AHS executes $\hat{a}_{t,1:K_t}$, updates $\mathcal{H}_{t+K_t}$, and replans from the next observation, using longer prefixes in stable phases and shorter ones near contact or uncertainty.

Posterior update with kernel forgetting. After executing the selected prefix, AHS uses the soft feedback $y_t = \sum_{k \in \mathcal{K}} \mu_t(k) q_{\text{mix},t}(k)$. Feedback is shared among nearby horizons through a Gaussian kernel centered at K_t , $w_t(k) \propto \exp[-(k - K_t)^2 / (2\sigma_K^2)]$, normalized over $\mathcal{K}$. The Beta state is updated with exponential forgetting [17]:

$$a_{t+1}(k) = \rho_f a_t(k) + \eta w_t(k) y_t, \quad (11)$$

$$b_{t+1}(k) = \rho_f b_t(k) + \eta w_t(k) (1 - y_t), \quad (12)$$

where ρ_f , η , and σ_K control forgetting, update strength, and neighborhood sharing. Forgetting lets AHS track phase-dependent horizon preferences, e.g., longer prefixes during free-space motion and shorter ones near contact.

3.2 Training-Time Query-Based Horizon Adapter

We further train a lightweight Query-based Horizon Adapter (QHA) to distill the AHS-derived preferences into a learned dense posterior $p_\phi(K_t = h \mid \mathbf{C}_t, \mathbf{Z}_t^A)$ over every action step $h \in \{1, \dots, H\}$, using VLM context tokens $\mathbf{C}_t$ and action-latent tokens $\mathbf{Z}_t^A$. Unlike AHS, which explicitly scores a sparse candidate set $\mathcal{K}$ at deployment, QHA predicts a per-step horizon distribution in one forward pass. As shown in Fig. 3, QHA uses learned horizon queries with bridge cross-attention over VLM context and action-latent tokens. Implementation details are in Appendix A.6.

Table 2: **Compact per-task success rates on RoboCasa GR1 Tabletop.** We show the Plate task group and 24-task Average. The full table appears in Appendix Tab. 6. Yellow columns use AHS and parentheses report percentage-point deltas over Base.

Task	QwenFAST +Qwen3VL	QwenPI +Qwen3VL	Isaac-GR00T N1.5	Isaac-GR00T N1.5 (AHS)	Isaac-GR00T N1.6	Isaac-GR00T N1.6 (AHS)	Qwen3GR00T +Qwen3VL	Qwen3GR00T +Qwen3VL (AHS)
PnPNovelFromPlateToBowl	52.0	52.0	58.0	58.0 (+0.0)	57.0	58.0 (+1.0)	60.0	60.0 (+0.0)
PnPNovelFromPlateToCardboardbox	30.0	40.0	40.0	48.0 (+8.0)	43.5	58.0 (+14.5)	50.0	54.0 (+4.0)
PnPNovelFromPlateToPan	48.0	36.0	44.0	48.0 (+4.0)	51.0	68.0 (+17.0)	54.0	54.0 (+0.0)
PnPNovelFromPlateToPlate	50.0	48.0	66.0	74.0 (+8.0)	78.7	82.0 (+3.3)	70.0	74.0 (+4.0)
Average	39.0	43.9	43.3	44.9 (+1.7)	47.6	51.4 (+3.8)	47.8	57.5 (+9.7)

During QHA training, each sample is supervised by a dense teacher distribution $\mu_t^*(h)$ over every action step:

$$\mu_t^*(h) \geq 0, \quad \sum_{h=1}^H \mu_t^*(h) = 1, \quad h = 1, \dots, H. \quad (13)$$

QHA minimizes $\mathcal{L}_{\text{QHA}} = \text{KL}(\mu_t^* \| p_\phi)$ with the base policy frozen. At deployment, the QHA posterior is used as a dense prior for AHS, while AHS supplies sparse online scores. We linearly interpolate the AHS runtime scores from $\mathcal{K}$ to the full horizon grid, $\tilde{s}_t(h) = \text{Interp}_{\mathcal{K} \rightarrow \{1, \dots, H\}}(\{s_t(k)\}_{k \in \mathcal{K}})_h$, and combine

$$s_{\text{eff},t}(h) = \tilde{s}_t(h) p_\phi(K_t = h \mid \mathbf{C}_t, \mathbf{Z}_t^A)^\gamma, \quad h = 1, \dots, H, \quad (14)$$

where $\gamma \geq 0$ controls the prior strength. Expected-round selection is then applied on the dense score, yielding the hybrid AHS+QHA setting while allowing AHS to use amortized knowledge in familiar phases and remain responsive to online evidence under episode-specific deviations.

4 Experiments

We evaluate the proposed horizon-aware execution strategy on two simulation benchmarks: RoboTwin2.0 [18] and RoboCasa GR1 Tabletop [19]. We report success rates (%) by comparing each base policy with AHS and, when available, AHS+QHA. For every baseline, AHS uses only the action-expert evidence in Sec. 2, and no policy weights are modified.

4.1 RoboTwin2.0

RoboTwin2.0 [18] contains 50 bimanual manipulation tasks. We select 8 representative tasks covering a range of difficulty levels, each evaluated under a clean (Easy) and a randomized (Hard) setting with 100 rollouts per condition. The main table evaluates three base policies: π_0 [1], $\pi_{0.5}$ [2], and Fast-WAM [7]. These base policies are fine-tuned on the RoboTwin2.0 training set (see Appendix A for checkpoint details) and are evaluated in a task-specific fine-tuned regime rather than zero-shot. We compare the vanilla baselines under their default evaluation settings against their AHS-augmented counterparts, using $\mathcal{K} = \{10, 20, 30, 40, 50\}$ for π_0 and $\pi_{0.5}$, and $\mathcal{K} = \{10, 20, 30\}$ for Fast-WAM.

Tab. 1 reports Base, AHS, and AHS+QHA for π_0 and $\pi_{0.5}$, and Base and AHS for Fast-WAM. AHS improves the overall success rate for all three policies in the main table: π_0 (18.2 $\rightarrow$ 21.4, +3.2), $\pi_{0.5}$ (29.6 $\rightarrow$ 36.2, +6.6), and Fast-WAM (86.8 $\rightarrow$ 88.1, +1.3). The hybrid setting improves π_0 (18.2 $\rightarrow$ 24.9, +6.7) and $\pi_{0.5}$ (29.6 $\rightarrow$ 39.1, +9.5). The strongest per-task gain is Handover Mic Hard for π_0 with AHS+QHA (2 $\rightarrow$ 30, +28). Stronger baselines yield smaller aggregate changes: Fast-WAM also improves on Hard (85.2 $\rightarrow$ 87.4, +2.1), despite its high base success rate.

4.2 RoboCasa GR1 Tabletop

RoboCasa GR1 Tabletop [19] provides 24 diverse pick-and-place tasks spanning everyday objects, receptacles, and novel source–target combinations, with 50 rollouts per task. We apply AHS to three GR00T-family backbones: Isaac-GR00T N1.5 [4], Isaac-GR00T N1.6 [4], and Qwen3GR00T [5, 6]. N1.5 and N1.6 are evaluated zero-shot on RoboCasa, while Qwen3GR00T is fine-tuned from the StarVLA codebase [5, 6] (see Appendix A for details). We also include two StarVLA-codebase references [5, 6] without AHS: QwenFAST (discrete tokens) and QwenPI (flow-matching action expert), both using Qwen3VL [20]. For each backbone, we report the original evaluation (Base) and the AHS-selected horizon policy, with the same candidate set $\mathcal{K} = \{4, 8, 12, 16\}$.

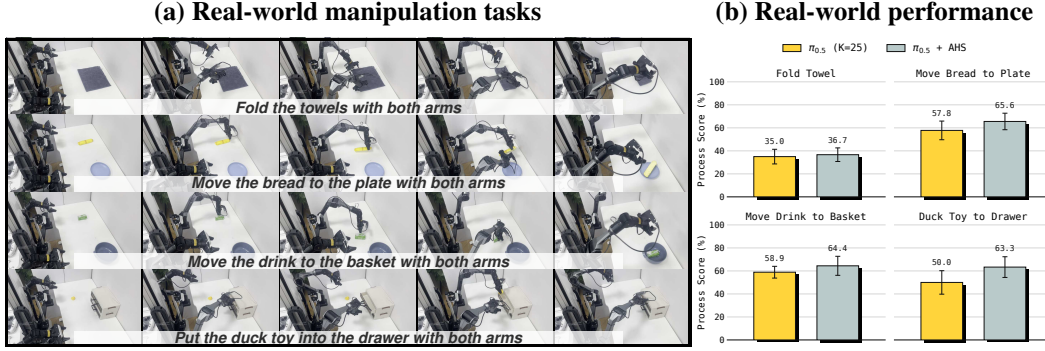


Figure 4: **Real-world deployment.** (a) Four bimanual household tasks: fold towels, move bread to a plate, move a drink to a basket, and put a duck toy into a drawer. (b) Normalized process scores for $\pi_{0.5} (K = 25)$ and $\pi_{0.5} + \text{AHS}$ over 15 rollouts per task.

Table 3: **Candidate-set scaling on $\pi_{0.5}$.** Success and per-replan overhead over four RoboTwin2.0 tasks as $|\mathcal{K}|$ increases. Fixed denotes $\pi_{0.5}$ without horizon selection ($K = 50$).

Candidate set	$ \mathcal{K} $	Success (%)	$\Delta\text{Success}$	Replan time (ms)	Overhead (%)	Replan counts	Ep. time (s)
Fixed ($K=50$)	—	22.00	—	95.83	—	4.22	22.43
$\mathcal{K}_5$	5	33.00	+11.00	96.50	0.70	8.36	23.17
$\mathcal{K}_{10}$	10	32.50	+10.50	96.84	1.05	9.34	23.08
$\mathcal{K}_{50}$	50	34.88	+12.88	99.28	3.60	10.57	21.39

Tab. 2 shows a compact per-task view, with the full 24-task breakdown in Appendix Tab. 6. AHS improves the average success rate for all three backbones: N1.5 ($43.3 \rightarrow 44.9$, +1.7), N1.6 ($47.6 \rightarrow 51.4$, +3.8), and Qwen3GR00T ($47.8 \rightarrow 57.5$, +9.7). Qwen3GR00T improves by double digits on 12 of 24 tasks and peaks on Tray to Tiered Shelf ($16.0 \rightarrow 44.0$, +28). Performance drops marginally on specific task instances, such as N1.5 on Potato to Microwave ($54.0 \rightarrow 36.0$, -18), but positive averages across all backbones indicate transfer across model scales and action experts.

4.3 Real-World Deployment

We evaluate real-world deployment with $\pi_{0.5}$ on the AgileX COBOT Magic ALOHA-style bimanual platform under perception noise, actuation delay, and contact-rich interactions. As shown in Fig. 4, the evaluation covers four household tasks: folding towels, moving bread to a plate, moving a drink to a basket, and placing a duck toy into a drawer. These tasks span free-space approach, contact, transport, alignment, and placement, where a fixed horizon may over-commit before feedback or replan unnecessarily during stable motion. We compare fixed $\pi_{0.5}$ with $K = 25$ against $\pi_{0.5} + \text{AHS}$. Reported scores average predefined sub-steps and rollouts, rather than binary success, to capture partial progress under limited trials, with criteria in Appendix A.1. Across the four tasks, $\pi_{0.5} + \text{AHS}$ improves the equal-task mean normalized process score from 50.4% to 57.5%, an absolute gain of +7.1 percentage points. The largest gain appears on Duck Toy to Drawer, whose opening, grasping, placement, and closing phases improve from 50.0% to 63.3% (+13.3 percentage points).

4.4 Ablation Studies

We conduct two controlled ablation studies to isolate the contribution of each design component. Both studies use the last four RoboTwin2.0 tasks (Place A2B Left, Place Bread Basket, Place Bread Skillet, Place Can Basket) under Easy and Hard settings with 100 rollouts per condition.

Candidate-set scaling. Tab. 3 reports the performance–overhead trade-off of AHS on $\pi_{0.5}$ as the candidate set grows from $|\mathcal{K}| = 5$ to $|\mathcal{K}| = 50$. The baseline is $\pi_{0.5}$ without horizon selection (fixed $K = 50$), which achieves 22.0% success rate over these four tasks. Using the coarsest candidate set $\mathcal{K}_5 = \{10, 20, 30, 40, 50\}$ increases the success rate to 33.0% (+11.0 percentage points) with only 0.70% extra per-replan overhead (0.67 ms added to the 95.83 ms fixed inference baseline). Increasing to $\mathcal{K}_{50} = \{1, 2, \dots, 50\}$ yields a further gain to 34.9% (+12.9 percentage points) at 3.60% overhead (3.45 ms). The episode-level cost remains modest: even with 50 candidates, the average

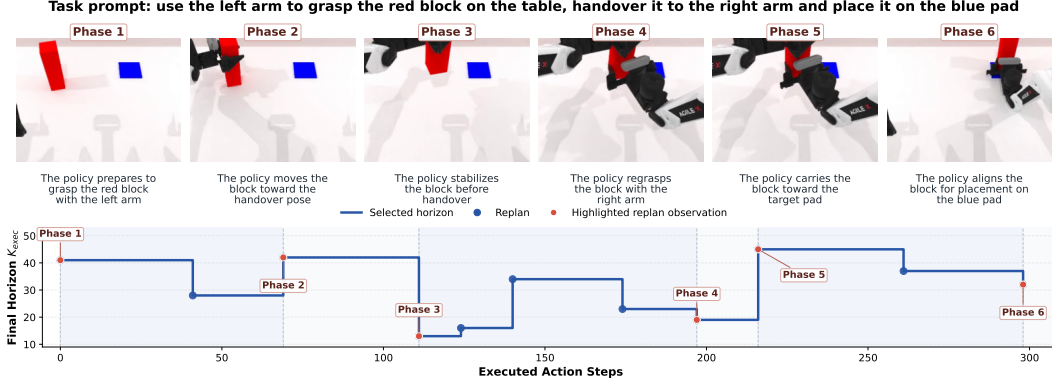


Figure 5: π_0 **Handover Block** case study. The selected execution horizon changes with task phase in a successful RoboTwin2.0 rollout.

successful episode completes in 21.4 s, comparable to the fixed baseline (22.4 s). Although adaptive execution increases the number of replans, these additional decisions are evidence-filtered and help avoid low-quality horizon commitments that could otherwise destabilize execution. These results confirm that AHS evidence fusion and horizon selection add negligible computation relative to policy inference, and that even a small candidate set accounts for most of the observed improvement.

Component ablation. Tab. 4 decomposes AHS into its constituent mechanisms on π_0 over the same four tasks, comparing six configurations: (i) Base π_0 under its default evaluation setting and no horizon selection, (ii) inter-chunk continuity evidence only, (iii) intra-chunk stability evidence only, (iv) both evidence terms with greedy expected-round selection and no posterior, (v) the full AHS with Beta posterior and kernel forgetting, and (vi) Full AHS+QHA.

Table 4: **Component ablation on π_0 .** Mean success over four RoboTwin2.0 tasks. Bold/underline mark best/second-best.

Method	Easy (%)	Hard (%)	Overall (%)
Base π_0 (default)	20.75	4.00	12.38
Inter-chunk only	23.50	4.00	13.75
Intra-chunk only	23.75	<u>5.25</u>	14.50
Both evidence terms, no posterior	<u>25.50</u>	3.75	14.63
Full AHS	24.25	5.75	15.00
Full AHS+QHA	27.50	4.00	15.75

The base policy obtains 12.4% overall. Each individual evidence term provides a modest gain (inter-only: 13.8%, intra-only: 14.5%), confirming that neither evidence term alone is sufficient. Combining both evidence terms without the posterior reaches 14.6% on the same four-task ablation subset. The full AHS with Beta posterior and kernel forgetting reaches 15.0% overall, with the largest Hard-setting gain from temporal smoothing. AHS+QHA obtains the highest overall success rate (15.8%), showing that QHA distills useful AHS-derived preferences.

Case study. Fig. 5 illustrates why adaptive horizons are useful. In the π_0 Handover Block rollout, AHS selects shorter horizons during grasping and transport phases, where closed-loop corrections matter, and longer horizons after contact, where grasp-and-transport becomes stable. This phase-dependent behavior matches the role of the online Beta posterior with kernel forgetting in Sec. 3.1.

5 Conclusion

We presented a horizon-aware execution framework for flow-based robot foundation policies, grounded in the view that the execution horizon is a latent variable of the action expert rather than a scalar chunk-size heuristic. AHS uses intra-chunk spectral stability and inter-chunk continuity as training-free evidence for an online Beta posterior with kernel forgetting, while QHA distills these preferences into an amortized posterior that can be fused with online evidence at deployment. Empirically, AHS improves success rates across multiple base policies and two simulation benchmarks, transfers to real-world bimanual manipulation, and incurs only 0.7–3.6% per-replan overhead. Ablations show that both evidence terms, the temporal posterior, and the hybrid AHS+QHA setting contribute to performance. Future work includes multi-chunk horizon inference, cross-policy transfer of QHA dense priors, and broader real-world validation across hardware and domain shifts.

References

- [1] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0 : A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.
- [2] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, et al. $\pi_{0.5}$: A vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025.
- [3] Jinliang Zheng, Jianxiong Li, Zhihao Wang, Dongxiu Liu, Xirui Kang, Yuchun Feng, Yinan Zheng, Jiayin Zou, Yilun Chen, Jia Zeng, Tai Wang, Ya-Qin Zhang, Jingjing Liu, and Xianyu Zhan. X-VLA: Soft-prompted transformer as scalable cross-embodiment vision-language-action model. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=kt51kZH4aG>.
- [4] Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, et al. Gr00t n1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734*, 2025.
- [5] Jinhui Ye, Ning Gao, Senqiao Yang, Jinliang Zheng, Zixuan Wang, Yuxin Chen, Pengguang Chen, Yilun Chen, Shu Liu, and Jiaya Jia. Starvla- α : Reducing complexity in vision-language-action systems. *arXiv preprint arXiv:2604.11757*, 2026.
- [6] StarVLA Community. Starvla: A lego-like codebase for vision-language-action model development. *arXiv preprint arXiv:2604.05014*, 2026.
- [7] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-wam: Do world action models need test-time future imagination? *arXiv preprint arXiv:2603.16666*, 2026.
- [8] Hongzhe Bi, Hengkai Tan, Shenghao Xie, Zeyuan Wang, Shuhe Huang, Haitian Liu, Ruowen Zhao, Yao Feng, Chendong Xiang, Yinze Rong, et al. Motus: A unified latent action world model. *arXiv preprint arXiv:2512.13030*, 2025.
- [9] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, et al. World action models are zero-shot policies. In *ICLR 2026 the 2nd Workshop on World Models: Understanding, Modelling and Scaling*.
- [10] Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 44(10-11):1684–1704, 2025.
- [11] Yuejiang Liu, Jubayer Ibn Hamid, Annie Xie, Yoonho Lee, Max Du, and Chelsea Finn. Bidirectional decoding: Improving action chunking via guided test-time sampling. In *International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=qZmn2hkuzw>.
- [12] Yuanchang Liang, Xiaobo Wang, Kai Wang, Shuo Wang, Xiaojiang Peng, Haoyu Chen, David Kim Huat Chua, and Prahlad Vadakkepat. Adaptive action chunking at inference-time for vision-language-action models. *arXiv preprint arXiv:2604.04161*, 2026.
- [13] Zipeng Fu, Tony Z. Zhao, and Chelsea Finn. Mobile ALOHA: Learning bimanual mobile manipulation using low-cost whole-body teleoperation. In *8th Annual Conference on Robot Learning*, 2024. URL <https://openreview.net/forum?id=F06tePGRZj>.
- [14] Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *The Eleventh International Conference on Learning Representations*.
- [15] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.

- [16] Daniel J Russo, Benjamin Van Roy, Abbas Kazerouni, Ian Osband, and Zheng Wen. A tutorial on thompson sampling. *Foundations and Trends in Machine Learning*, 11(1):1–96, 2018.
- [17] Vishnu Raj and Sheetal Kalyani. Tuning non-stationary thompson sampling. In *International Conference on Communication Systems and Networks (COMSNETS)*, 2017.
- [18] Tianxing Chen, Zanzin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Qiwei Liang, Zixuan Li, Xianliang Lin, Yiheng Ge, Zhenyu Gu, et al. Robotwin 2.0: A scalable data generator and benchmark with strong domain randomization for robust bimanual robotic manipulation. *arXiv preprint arXiv:2506.18088*, 2025.
- [19] Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. Robocasa: Large-scale simulation of everyday tasks for generalist robots. In *RSS 2024 Workshop: Data Generation for Robotics*, 2024.
- [20] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, et al. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025.
- [21] Zipeng Fu, Tony Z Zhao, and Chelsea Finn. Mobile aloha: Learning bimanual mobile manipulation using low-cost whole-body teleoperation. In *8th Annual Conference on Robot Learning*, 2024.
- [22] Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- [23] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- [24] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Tomas Jackson, Sally Jesmonth, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Kuang-Huei Lee, Sergey Levine, Yao Lu, Utsav Malla, Deeksha Manjunath, Igor Mordatch, Ofir Nachum, Carolina Parada, Jodilyn Peralta, Emily Perez, Karl Pertsch, Jornell Quiambao, Kanishka Rao, Michael S Ryoo, Grecia Salazar, Pannag R Sanketi, Kevin Sayed, Jaspiar Singh, Sumedh Sontakke, Austin Stone, Clayton Tan, Huong Tran, Vincent Vanhoucke, Steve Vega, Quan H Vuong, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and Brianna Zitkovich. RT-1: Robotics Transformer for Real-World Control at Scale. In *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea, July 2023. doi: 10.15607/RSS.2023.XIX.025.
- [25] Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S. Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu, Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov, Ryan Julian, Nikhil J. Joshi, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog, Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In Jie Tan, Marc Toussaint, and Kourosh Darvish, editors, *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 2165–2183. PMLR, 06–09 Nov 2023. URL <https://proceedings.mlr.press/v229/zitkovich23a.html>.
- [26] Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, Wenlong Huang, Yevgen Chebotar, Pierre Sermanet, Daniel Duckworth, Sergey Levine, Vincent Vanhoucke, Karol Hausman, Marc

- 403 Toussaint, Klaus Greff, Andy Zeng, Igor Mordatch, and Pete Florence. PaLM-e: An embodied
404 multimodal language model. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara
405 Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International
406 Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*,
407 pages 8469–8488. PMLR, 23–29 Jul 2023. URL [https://proceedings.mlr.press/v202/
408 driess23a.html](https://proceedings.mlr.press/v202/driess23a.html).
- 409 [27] Abby O’Neill, Abdul Rehman, Abhiram Maddukuri, Abhishek Gupta, Abhishek Padalkar,
410 Abraham Lee, Acorn Pooley, Agrim Gupta, Ajay Mandlekar, Ajinkya Jain, et al. Open x-
411 embodiment: Robotic learning datasets and rt-x models: Open x-embodiment collaboration 0.
412 In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 6892–6903.
413 IEEE, 2024.
- 414 [28] Dibya Ghosh, Homer Rich Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey
415 Hejna, Tobias Kreiman, Charles Xu, Jianlan Luo, You Liang Tan, Lawrence Yunliang Chen,
416 Quan Vuong, Ted Xiao, Pannag R Sanketi, Dorsa Sadigh, Chelsea Finn, and Sergey Levine.
417 Octo: An Open-Source Generalist Robot Policy. In *Proceedings of Robotics: Science and
418 Systems*, Delft, Netherlands, July 2024. doi: 10.15607/RSS.2024.XX.090.
- 419 [29] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair,
420 Rafael Rafailov, Ethan P Foster, Pannag R Sanketi, Quan Vuong, Thomas Kollar, Benjamin
421 Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. Open-
422 VLA: An open-source vision-language-action model. In *8th Annual Conference on Robot
423 Learning*, 2024. URL <https://openreview.net/forum?id=ZMnD6QZAE6>.
- 424 [30] BAAI RoboBrain Team, Mingyu Cao, Huajie Tan, Yuheng Ji, Xiansheng Chen, Minglan Lin,
425 Zhiyu Li, Zhou Cao, Pengwei Wang, Enshen Zhou, et al. Robobrain 2.0 technical report. *arXiv
426 preprint arXiv:2507.02029*, 2025.
- 427 [31] Junjie Wen, Yichen Zhu, Jinming Li, Zhibin Tang, Chaomin Shen, and Feifei Feng. Dexvla:
428 Vision-language model with plug-in diffusion expert for general robot control. In *9th Annual
429 Conference on Robot Learning*.
- 430 [32] Songming Liu, Lingxuan Wu, Bangguo Li, Hengkai Tan, Huayu Chen, Zhengyi Wang, Ke Xu,
431 Hang Su, and Jun Zhu. RDT-1b: a diffusion foundation model for bimanual manipulation.
432 In *The Thirteenth International Conference on Learning Representations*, 2025. URL [https:
433 //openreview.net/forum?id=yAzN4tz7oI](https://openreview.net/forum?id=yAzN4tz7oI).
- 434 [33] Dong Jing, Gang Wang, Jiaqi Liu, Weiliang Tang, Zelong Sun, Yunchao Yao, Zhenyu Wei,
435 Yunhui Liu, Zhiwu Lu, and Mingyu Ding. Mixture of horizons in action chunking. *arXiv
436 preprint arXiv:2511.19433*, 2025.
- 437 [34] Haoxuan Wang, Gengyu Zhang, Yan Yan, Ramana Rao Kompella, and Gaowen Liu. Vla knows
438 its limits. *arXiv preprint arXiv:2602.21445*, 2026.
- 439 [35] Jiyao Zhang, Zimu Han, Junhan Wang, Xionghao Wu, Shihong Lin, Jinzhou Li, Hongwei
440 Fan, Ruihai Wu, Dongjiang Li, and Hao Dong. Hipolicy: Hierarchical multi-frequency action
441 chunking for policy learning. *arXiv preprint arXiv:2604.06067*, 2026.
- 442 [36] Junhyuk So, Chiwoong Lee, Shinyoung Lee, Jungseul Ok, and Eunhyeok Park. Improving
443 generative behavior cloning via self-guidance and adaptive chunking. In *The Thirty-ninth Annual
444 Conference on Neural Information Processing Systems*, 2026. URL [https://openreview.
445 net/forum?id=GctsZXLcPl](https://openreview.net/forum?id=GctsZXLcPl).

How Far Can Robots Trust Action Chunks? Horizon Adaptation via Action-Expert Evidence

Appendix

A Experimental Setup

This section provides full details for reproducing the experimental results in Sec. 4.

A.1 Real-world Experiments

Hardware Setup. We conduct real-world experiments on an AgileX Cobot Magic platform configured as an ALOHA-style bimanual system [13, 21], as shown in Fig. 6. The platform consists of four 6-DoF Piper arms, with two leader arms used for human teleoperation and two follower arms used for data collection and autonomous policy execution. The perception system includes three RealSense D435 cameras: one front-view camera and two wrist-mounted cameras, one on each follower arm.

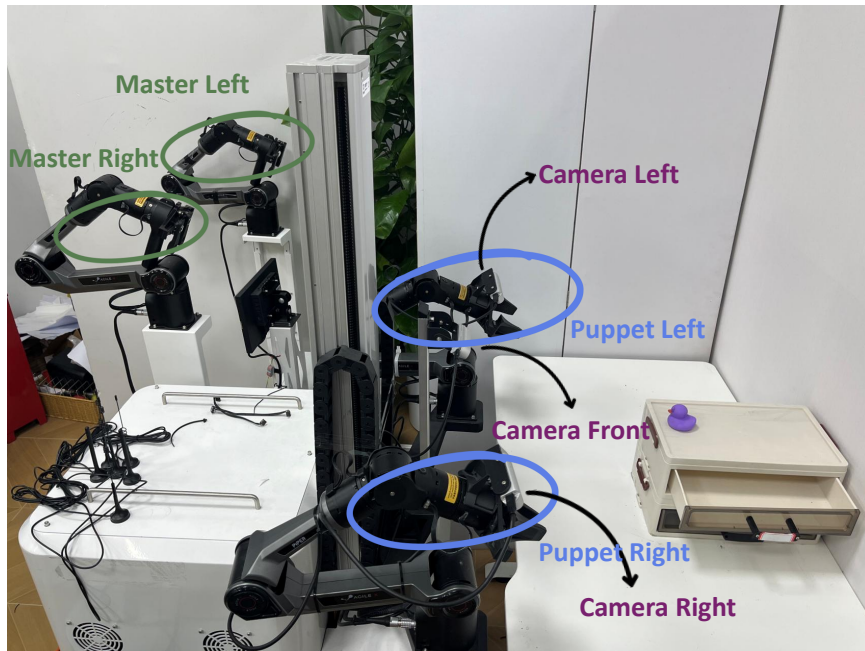


Figure 6: **Real-world robot platform.** AgileX Cobot Magic configured as an ALOHA-style bimanual system for teleoperation, data collection, and autonomous policy execution.

Tasks. We evaluate four bimanual household manipulation tasks on the real robot, as visualized in Fig. 4: Fold Towel, Move Bread to Plate, Move Drink to Basket, and Put Duck Toy into Drawer. The tasks are chosen to cover complementary phases where the executable horizon should change, including free-space approach, contact, transport, alignment, and placement. The language instructions are: “Fold the towel with both arms,” “Move the bread to the plate with both arms,” “Move the drink to the basket with both arms,” and “Put the duck toy into the drawer with both arms.” For each task, we collect 200 human-teleoperated demonstrations covering tabletop object-position generalization.

Evaluation. We evaluate each real-world task with a fixed number of rollouts: 15 trials per task. All real-robot data is recorded at 30 FPS. During evaluation, the tabletop positions of all task-relevant objects are varied across rollouts to test spatial generalization rather than replaying a single fixed layout. The deployed base policy is $\pi_{0.5}$, and we compare fixed-prefix execution with $K = 25$ against $\pi_{0.5} + \text{AHS}$. Unlike simulation, where task completion can often be summarized by a binary success label, our real-world tasks contain multiple manipulation phases whose partial progress is meaningful

under a limited number of physical trials. We therefore use fine-grained process-level evaluation metrics. Each task is decomposed into key sub-steps, each sub-step is assigned a score according to predefined criteria, and the final task score is the normalized average across sub-steps and rollouts. Unless otherwise specified, each sub-step receives 0 points for failure, 0.5 points for recovered or imperfect completion, and 1 point for smooth and accurate completion. The scoring criteria are as follows.

1. Move Bread to Plate.

- *Step 1: Left-hand grasping the bread.*
 - 0 point: The robot fails to grasp the bread.
 - 0.5 point: The robot grasps the bread after multiple attempts.
 - 1 point: The robot grasps the bread smoothly on the first attempt.
- *Step 2: Bimanual handover.*
 - 0 point: The handover fails.
 - 0.5 point: The handover succeeds but the receiving grasp is unstable or awkward.
 - 1 point: The handover succeeds with a stable and well-aligned grasp.
- *Step 3: Placing the bread on the plate.*
 - 0 point: The bread is not placed on the plate.
 - 0.5 point: The bread is placed on the plate but is poorly aligned or roughly placed.
 - 1 point: The bread is placed cleanly on the plate.

2. Move Drink to Basket.

- *Step 1: Left-hand pushing the drink.*
 - 0 point: The robot only tilts the drink without useful displacement.
 - 0.5 point: The robot moves the drink to an acceptable position but leaves noticeable tilt or misalignment.
 - 1 point: The robot pushes the drink to a good position with stable alignment.
- *Step 2: Right-hand grasping the drink.*
 - 0 point: The robot fails to grasp the drink.
 - 0.5 point: The robot grasps the drink after multiple attempts.
 - 1 point: The robot grasps the drink smoothly on the first attempt.
- *Step 3: Placing the drink into the basket.*
 - 0 point: Placement fails or the drink is dropped.
 - 0.5 point: The drink is placed into the basket but roughly or with a collision.
 - 1 point: The drink is placed cleanly into the basket.

3. Fold Towel.

- *Step 1: First folding motion.*
 - 0 point: The towel is not folded over.
 - 0.5 point: The towel is folded over but misaligned.
 - 1 point: The towel is folded over with good alignment.
- *Step 2: Second folding motion.*
 - 0 point: The towel is not folded over.
 - 0.5 point: The towel is folded over but misaligned.
 - 1 point: The towel is folded over with good alignment.

4. Put Duck Toy into Drawer. The arm assignment is not fixed: the robot must infer which arm should open or close the drawer and which arm should manipulate the toy from the current tabletop layout.

- *Step 1: Opening the drawer with one arm.*
 - 0 point: The robot fails to open the drawer.
 - 0.5 point: The robot opens the drawer after multiple attempts.
 - 1 point: The robot opens the drawer smoothly on the first attempt.
- *Step 2: Grasping the duck toy with the other arm.*
 - 0 point: The robot fails to grasp the duck toy.
 - 0.5 point: The robot grasps the duck toy after multiple attempts.
 - 1 point: The robot grasps the duck toy smoothly on the first attempt.

- 523 - *Step 3: Placing the duck toy into the drawer.*
- 524 – 0 point: The robot fails to place the toy into the drawer.
- 525 – 0.5 point: The robot places the toy into the drawer after multiple attempts.
- 526 – 1 point: The robot places the toy into the drawer smoothly on the first attempt.
- 527 - *Step 4: Closing the drawer with one arm.*
- 528 – 0 point: The robot fails to close the drawer.
- 529 – 0.5 point: The robot closes the drawer after multiple attempts.
- 530 – 1 point: The robot closes the drawer smoothly on the first attempt.

531 A.2 Simulation Experiments

532 **RoboTwin2.0** [18] contains 50 bimanual manipulation tasks with strong domain randomization. We
 533 select 8 representative tasks covering a range of difficulty levels: Blocks Ranking RGB, Handover
 534 Block, Handover Mic, Hanging Mug, Place A2B Left, Place Bread Basket, Place Bread Skillet, and
 535 Place Can Basket. Each task is evaluated under two settings: *Easy* (clean, deterministic initial states)
 536 and *Hard* (randomized object poses, lighting, and distractor placement). Results report success rate
 537 over 100 rollouts per task per setting.

538 **RoboCasa GR1 Tabletop** [19] provides 24 pick and-place tasks spanning everyday object categories
 539 and novel source–target combinations. We evaluate all 24 tasks, with 50 rollouts per task. Task
 540 success is determined by the native RoboCasa success checker.

541 A.3 Base Policy Checkpoints

542 RoboTwin2.0 checkpoints follow the evaluation protocol of each base policy:

- 543 • π_0 [1]: we train LoRA adapters [22] with the released RoboTwin2.0 fine-tuning protocol.
 544 For each evaluated task, one adapter is trained on that task’s clean50 demonstrations and
 545 then evaluated. The action chunk size is $H = 50$, with a diffusion-based action expert [15]
 546 using 10 denoising steps.
- 547 • $\pi_{0.5}$ [2]: we use the official RoboTwin2.0 fine-tuning recipe for full-parameter fine-tuning.
 548 For each evaluated task, one task-specific checkpoint is trained on clean50 demonstra-
 549 tions and then evaluated. The action chunk size is $H = 50$, with a flow-matching action
 550 expert [14].
- 551 • X-VLA [3]: we evaluate the released X-VLA RoboTwin2.0 checkpoint. The action chunk
 552 size is $H = 30$, with a flow-matching action expert.
- 553 • Fast-WAM [7]: we evaluate the released Fast-WAM RoboTwin2.0 checkpoint. The action
 554 chunk size is $H = 32$, with a flow-matching action expert.

555 For RoboCasa GR1 Tabletop, all base models are evaluated from released checkpoints: Isaac-
 556 GR00T N1.5 and Isaac-GR00T N1.6 [4], and Qwen3GR00T from the StarVLA codebase [5, 6].

557 A.4 AHS Hyperparameter Configurations

558 AHS uses base-model-dependent candidate sets and continuity windows, with the remaining hyperpa-
 559 rameters shared unless noted:

- 560 • RoboTwin2.0 with π_0 and $\pi_{0.5}$: $\mathcal{K} = \{10, 20, 30, 40, 50\}$.
- 561 • RoboTwin2.0 with X-VLA and Fast-WAM: $\mathcal{K} = \{10, 20, 30\}$.
- 562 • RoboCasa GR1 Tabletop for all evaluated base models: $\mathcal{K} = \{4, 8, 12, 16\}$.
- 563 • Candidate-set scaling ablation: $\mathcal{K}_5 = \{10, 20, 30, 40, 50\}$, $\mathcal{K}_{10} = \{5, 10, 15, \dots, 50\}$, and
 564 $\mathcal{K}_{50} = \{1, 2, \dots, 50\}$.
- 565 • Intra-chunk penalty $\alpha_{\text{intra}} = 4.0$, inter-chunk penalty $\beta_{\text{inter}} = 4.0$.
- 566 • Fusion weight $\lambda = 0.5$ (equal contribution from intra- and inter- chunk evidence terms).
- 567 • Beta prior: $a_0(k) = b_0(k) = 1$ (uniform).
- 568 • Kernel forgetting rate $\rho = 0.99$, update strength $\eta = 1.0$, kernel bandwidth $\sigma = 10.0$.

- Expected-round selection with temperature $T_{\text{sel}} = 0.8$, Thompson sampling with $\epsilon = 0.05$ for exploration.
- Continuity window on RoboTwin2.0 with π_0 and $\pi_{0.5}$: $W_h = 60$ steps.
- Continuity window on RoboTwin2.0 with X-VLA and Fast-WAM: $W_h = 40$ steps.
- Continuity window on RoboCasa GR1 Tabletop for all evaluated base models: $W_h = 20$ steps.
- Intra-chunk high-frequency energy tau reference window: $W_\tau = 3$, z -mode: window RMS. Spectral cutoff fraction $c_{\text{cut}} = 0.25$.

For spectral scoring, velocity prefixes of different candidate horizons are zero-padded to the full chunk length H before FFT so that frequency bins are comparable across k .

When the inter-chunk evidence is unavailable at the beginning of an episode (insufficient execution history), AHS falls back to intra-chunk-only scoring ($\lambda = 0$). All evidence computation runs on the same GPU as policy inference without requiring additional device transfers.

A.5 Evaluation Protocol

For every (base policy, task, setting) combination, we run a fixed number of rollouts with AHS enabled. Each rollout starts from the standard initial state distribution of the benchmark. AHS is reset to its prior ($a_0 = b_0 = 1$) at the beginning of each episode. No information is carried across episodes. Success is determined by the benchmark’s native success checker. We report the raw success rate (percentage of successful rollouts) without any statistical post-processing such as bootstrapping.

RoboTwin2.0 experiments: Fast-WAM inference runs on a single NVIDIA H100 (80 GB). All other policy inference (π_0 , $\pi_{0.5}$, X-VLA) runs on a single NVIDIA RTX 4090 (24 GB). RoboCasa experiments use an NVIDIA RTX 4090 GPU. All experiments use 30 FPS camera input and policy inference frequency matched to the camera frame rate.

A.6 QHA Training Configurations

Training inputs and dense teacher distribution. During training, the frozen base policy produces the same full action chunk and rollout evidence used by AHS, together with a dense teacher distribution $\mu_t^*(h)$ over $h = 1, \dots, H$. On RoboTwin2.0, we train QHA separately for π_0 and $\pi_{0.5}$. For each base policy, the eight evaluated tasks are jointly trained with a single QHA adapter shared across tasks, while the base action generator remains frozen. The distillation data are the same RoboTwin2.0 Aloha-AgileX clean-50 LeRobot datasets used to train the corresponding task policies. We use task-balanced mixed batches with global batch size 256, giving 32 samples per task per batch across the eight tasks. Inputs include the three ALOHA RGB streams, robot state, action windows, VLM context tokens, and action-latent tokens. The QHA action-query window uses 64 history steps and 50 future action steps. The per-replan training tuple is

$$\{\hat{\mathbf{A}}_t, \mathcal{F}_t, \mathbf{C}_t, \mathbf{M}_t, \mathbf{Z}_t^A, \mu_t^*\}. \quad (15)$$

Here $\mathbf{C}_t \in \mathbb{R}^{N_c \times d_c}$ are context tokens from the VLM backbone, $\mathbf{M}_t$ is the context mask, and $\mathbf{Z}_t^A \in \mathbb{R}^{H \times d_z}$ are action-latent tokens obtained from the final action chunk. The dense teacher distribution is generated online by routing each sample to the frozen task-specific AHS teacher checkpoint for that task, with dense horizon candidates $h = 1, \dots, 50$ and score-normalization temperature $T_{\text{teach}} = 1.0$.

Architecture. QHA predicts a categorical posterior over dense action-step horizons rather than only over the sparse AHS candidate set:

$$p_\phi(K_t = h \mid \mathbf{C}_t, \mathbf{Z}_t^A), \quad h \in \{1, \dots, H\}. \quad (16)$$

Context and action-latent tokens are projected into a shared hidden space:

$$\tilde{\mathbf{C}}_t = g_C(\mathbf{C}_t), \quad \tilde{\mathbf{Z}}_t^A = g_A(\mathbf{Z}_t^A), \quad (17)$$

and augmented with positional encodings. Following transformer attention blocks [23], QHA maintains Q learned query tokens $\mathbf{Q}^{(0)} \in \mathbb{R}^{Q \times d}$. Each of the L bridge layers first cross-attends from

the queries to the VLM context, then to the action-latent tokens, and finally refines the queries by self-attention:

$$\mathbf{Q}^{(l,c)} = \mathbf{Q}^{(l)} + \text{CrossAttn}(\mathbf{Q}^{(l)}, \tilde{\mathbf{C}}_t, \tilde{\mathbf{C}}_t; \mathbf{M}_t), \quad (18)$$

$$\mathbf{Q}^{(l,a)} = \mathbf{Q}^{(l,c)} + \text{CrossAttn}(\mathbf{Q}^{(l,c)}, \tilde{\mathbf{Z}}_t^A, \tilde{\mathbf{Z}}_t^A), \quad (19)$$

$$\mathbf{Q}^{(l+1)} = \mathbf{Q}^{(l,a)} + \text{SelfAttn}(\mathbf{Q}^{(l,a)}). \quad (20)$$

After L bridge layers, query pooling and an output MLP produce dense horizon logits $\alpha_t \in \mathbb{R}^H$, and the posterior is

$$p_\phi(K_t = h \mid \mathbf{C}_t, \mathbf{Z}_t^A) = \frac{\exp(\alpha_{t,h})}{\sum_{j=1}^H \exp(\alpha_{t,j})}, \quad h = 1, \dots, H. \quad (21)$$

This query formulation lets the horizon head attend jointly to task-level multimodal context and to the action expert’s latent representation, while remaining small relative to the backbone policy. In the joint QHA runs, we use dense candidates $1, \dots, 50$, $Q = 8$ learned queries, one bridge layer, hidden size $d = 256$, eight attention heads, dropout 0.0, and the fusion gate enabled.

Optimization. For QHA head training, the head is supervised by the dense teacher distribution μ_t^* and the base policy is frozen:

$$\mathcal{L}_{\text{QHA}} = \text{KL}(\mu_t^* \parallel p_\phi). \quad (22)$$

The KL loss uses weight 1.0 and numerical floor $\epsilon = 10^{-6}$.

$$\mathcal{L}_{\text{total}} = \lambda_{\text{QHA}} \mathcal{L}_{\text{QHA}}, \quad \nabla_{\theta} \mathcal{L}_{\text{total}} = 0. \quad (23)$$

We train in QHA-only mode: the base policy is frozen, the flow-matching loss is set to zero for optimization, and only parameters under the QHA head are updated. We use AdamW with global-norm clipping 1.0, batch size 256, 10000 optimizer steps, warmup-cosine learning rate from a peak of 5×10^{-5} to 5×10^{-6} after 1000 warmup steps, weight decay 10^{-10} , and EMA decay 0.99.

AHS+QHA deployment. The learned posterior p_ϕ serves as a dense prior for online AHS. At deployment, AHS produces sparse runtime scores $s_t(k)$ over $k \in \mathcal{K}$. We linearly interpolate these scores to the dense horizon grid, $\tilde{s}_t(h) = \text{Interp}_{\mathcal{K} \rightarrow \{1, \dots, H\}}(\{s_t(k)\}_{k \in \mathcal{K}})_h$, and combine them with the QHA posterior:

$$s_{\text{eff},t}(h) = \tilde{s}_t(h) p_\phi(K_t = h \mid \mathbf{C}_t, \mathbf{Z}_t^A)^\gamma, \quad h = 1, \dots, H. \quad (24)$$

We use hybrid prior exponent $\gamma = 1.0$ in all reported AHS+QHA evaluations. Expected-round selection is then applied on the dense score. This keeps AHS efficient at test time while allowing QHA to contribute a per-step horizon prior.

B Full Simulation Benchmark Results

B.1 RoboTwin2.0

Table 5 reports the X-VLA [3] RoboTwin2.0 per-task results under Easy and Hard settings. X-VLA uses $\mathcal{K} = \{10, 20, 30\}$ and is evaluated with the same 8 tasks and 100-rollout protocol as the main RoboTwin2.0 table. The main table reports Base, AHS, and AHS+QHA results for π_0 and $\pi_{0.5}$, and Base/AHS results for Fast-WAM. For X-VLA, AHS slightly improves overall success ($47.5 \rightarrow 47.8, +0.3$). Easy increases ($74.5 \rightarrow 75.8, +1.3$), while Hard decreases slightly ($20.4 \rightarrow 19.8, -0.6$).

B.2 RoboCasa GR1 Tabletop

Table 6 reports the full 24-task RoboCasa GR1 Tabletop breakdown. The main paper shows a compact view with one task group and the 24-task average.

C More Case Studies

Fig. 7 extends the main case study with additional π_0 RoboTwin2.0 rollouts. The examples cover both clean and randomized settings and illustrate that AHS adapts the execution horizon across task phases rather than using a fixed chunk length.

Table 5: **X-VLA per-task success rates on RoboTwin2.0.** Easy/Hard denote clean/randomized settings. Yellow columns use AHS and parentheses report percentage-point deltas over Base.

Task	X-VLA [3] Base		X-VLA [3] (AHS)	
	Easy	Hard	Easy	Hard
Blocks Ranking RGB	87	34	95 (+8)	38 (+4)
Handover Block	89	2	88 (-1)	1 (-1)
Handover Mic	97	1	100 (+3)	1 (+0)
Hanging Mug	32	8	35 (+3)	4 (-4)
Place A2B Left	31	25	40 (+9)	21 (-4)
Place Bread Basket	87	41	80 (-7)	42 (+1)
Place Bread Skillet	86	22	88 (+2)	23 (+1)
Place Can Basket	87	30	80 (-7)	28 (-2)
Average	74.5	20.4	75.8 (+1.3)	19.8 (-0.6)
Overall		47.5	47.8 (+0.3)	

Table 6: **Full per-task success rates on RoboCasa GR1 Tabletop.** Full 24-task breakdown. Yellow columns use AHS and parentheses report percentage-point deltas over Base.

Task	QwenFAST +Qwen3VL	QwenPI +Qwen3VL	Isaac-GR00T N1.5	Isaac-GR00T N1.5 (AHS)	Isaac-GR00T N1.6	Isaac-GR00T N1.6 (AHS)	Qwen3GR00T +Qwen3VL	Qwen3GR00T +Qwen3VL (AHS)
PnPbottleToCabinetClose	38.0	26.0	64.0	68.0 (+4.0)	51.5	54.0 (+2.5)	46.0	64.0 (+18.0)
PnPCanToDrawerClose	44.0	62.0	18.0	12.0 (-6.0)	13.0	12.0 (-1.0)	80.0	80.0 (+0.0)
PnPCupToDrawerClose	56.0	42.0	12.0	4.0 (-8.0)	8.5	14.0 (+5.5)	54.0	52.0 (-2.0)
PnPMilkToMicrowaveClose	44.0	50.0	38.0	34.0 (-4.0)	14.0	20.0 (+6.0)	48.0	42.0 (-6.0)
PnPPotatoToMicrowaveClose	14.0	42.0	54.0	36.0 (-18.0)	41.5	50.0 (+8.5)	28.0	28.0 (+0.0)
PnPWineToCabinetClose	14.0	32.0	16.0	20.0 (+4.0)	16.5	24.0 (+7.5)	46.0	52.0 (+6.0)
PnPNovelFromCuttingboardToBasket	54.0	40.0	50.0	52.0 (+2.0)	58.0	54.0 (-4.0)	48.0	70.0 (+22.0)
PnPNovelFromCuttingboardToCardboardbox	42.0	46.0	36.0	34.0 (-2.0)	46.5	46.0 (-0.5)	40.0	54.0 (+14.0)
PnPNovelFromCuttingboardToPan	58.0	60.0	68.0	64.0 (-4.0)	68.5	80.0 (+11.5)	68.0	80.0 (+12.0)
PnPNovelFromCuttingboardToPot	58.0	40.0	34.0	56.0 (+22.0)	65.0	64.0 (-1.0)	52.0	76.0 (+24.0)
PnPNovelFromCuttingboardToTieredbasket	40.0	44.0	46.0	32.0 (-14.0)	46.5	54.0 (+7.5)	56.0	44.0 (-12.0)
PnPNovelFromPlacematToBasket	36.0	44.0	50.0	46.0 (-4.0)	58.5	48.0 (-10.5)	42.0	54.0 (+12.0)
PnPNovelFromPlacematToBowl	38.0	52.0	50.0	62.0 (+12.0)	57.5	60.0 (+2.5)	44.0	66.0 (+22.0)
PnPNovelFromPlacematToPlate	42.0	50.0	62.0	66.0 (+4.0)	63.0	82.0 (+19.0)	48.0	72.0 (+24.0)
PnPNovelFromPlacematToTieredshelf	18.0	28.0	14.0	26.0 (+12.0)	28.5	36.0 (+7.5)	18.0	20.0 (+2.0)
PnPNovelFromPlateToBowl	52.0	52.0	58.0	58.0 (+0.0)	57.0	58.0 (+1.0)	60.0	60.0 (+0.0)
PnPNovelFromPlateToCardboardbox	30.0	40.0	40.0	48.0 (+8.0)	43.5	58.0 (+14.5)	50.0	54.0 (+4.0)
PnPNovelFromPlateToPan	48.0	36.0	44.0	48.0 (+4.0)	51.0	68.0 (+17.0)	54.0	54.0 (+0.0)
PnPNovelFromPlateToPlate	50.0	48.0	66.0	74.0 (+8.0)	78.7	82.0 (+3.3)	70.0	74.0 (+4.0)
PnPNovelFromTrayToCardboardbox	28.0	34.0	44.0	52.0 (+8.0)	51.5	48.0 (-3.5)	38.0	56.0 (+18.0)
PnPNovelFromTrayToPlate	34.0	64.0	50.0	60.0 (+10.0)	71.0	68.0 (-3.0)	56.0	62.0 (+6.0)
PnPNovelFromTrayToPot	46.0	44.0	46.0	50.0 (+4.0)	64.5	64.0 (-0.5)	50.0	66.0 (+16.0)
PnPNovelFromTrayToTieredbasket	36.0	50.0	44.0	38.0 (-6.0)	57.0	56.0 (-1.0)	36.0	56.0 (+20.0)
PnPNovelFromTrayToTieredshelf	16.0	28.0	34.0	38.0 (+4.0)	31.5	34.0 (+2.5)	16.0	44.0 (+28.0)
Average	39.0	43.9	43.3	44.9 (+1.7)	47.6	51.4 (+3.8)	47.8	57.5 (+9.7)

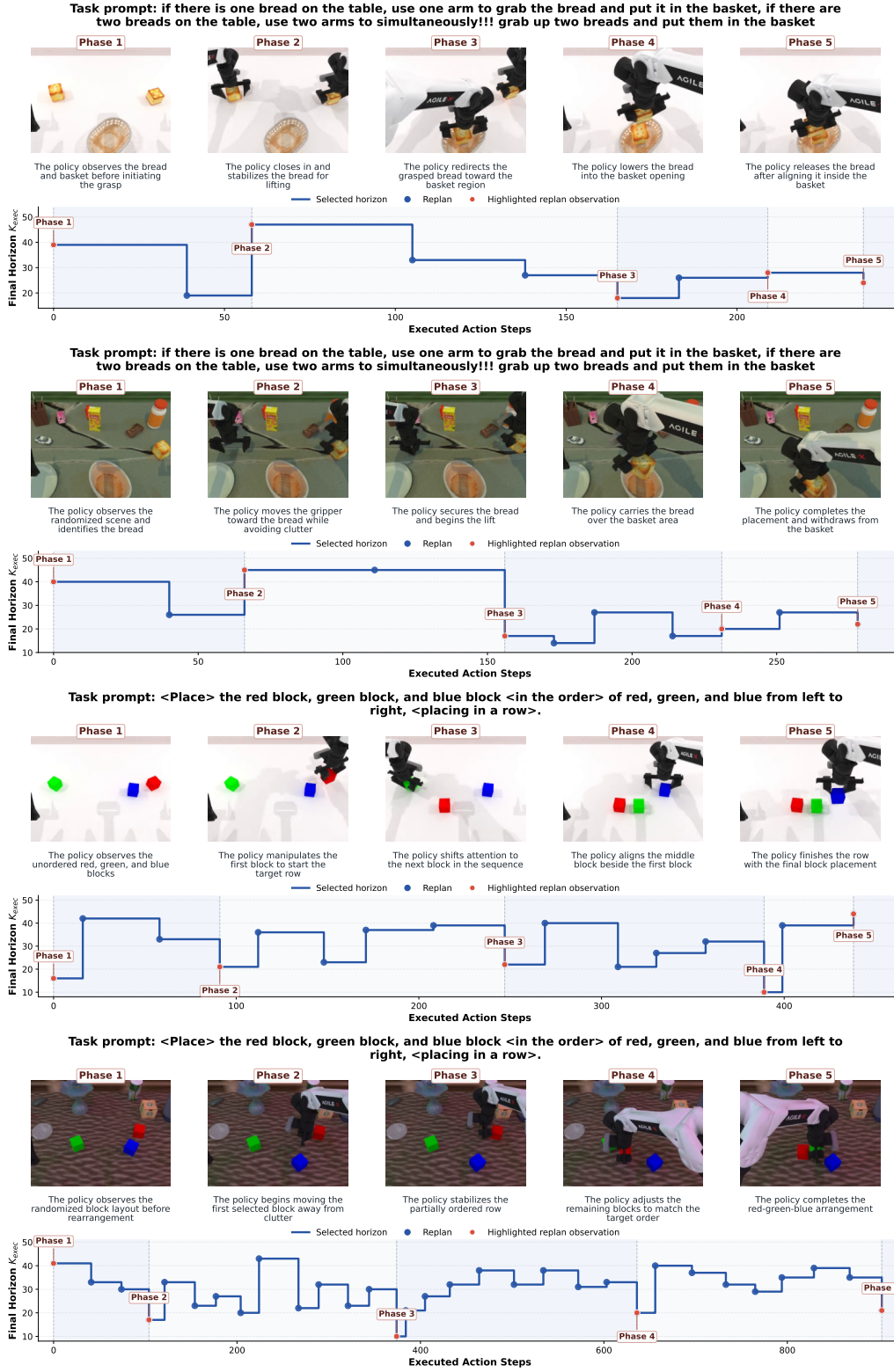


Figure 7: **Additional π_0 case studies on RoboTwin2.0.** Four successful rollouts show phase-dependent AHS horizons across Place Bread Basket and Blocks Ranking RGB under Easy and Hard settings.

D Related Work

Robot foundation policies and vision-language-action models. Large-scale robot policies increasingly combine broad visual-language pretraining with robot trajectory data. RT-1 showed that transformer policies can scale with diverse real-robot data [24], while RT-2 and PaLM-E connected web-scale vision-language knowledge to embodied control [25, 26]. Open X-Embodiment and Octo further emphasized cross-robot data standardization and generalist policy initialization [27, 28], and OpenVLA made this direction more accessible through an open 7B VLA model [29]. Embodied foundation models such as RoboBrain 2.0 broaden generalist robot reasoning and control interfaces [30]. Recent robot foundation policies such as π_0 , $\pi_{0.5}$, X-VLA, DexVLA, Isaac-GR00T, StarVLA, and Fast-WAM extend these ideas with flow-matching or diffusion action experts, cross-embodiment prompting, humanoid generalist pretraining, and world-action modeling [1–7, 31]. Video- and world-model-based policies further couple action prediction with future-state prediction or imagination [7–9]. Our work is orthogonal to these backbone designs: it treats the action expert as a frozen generator and adapts how much of each generated chunk should be executed before re-observation.

Generative action policies and action chunking. Action chunking improves temporal consistency by predicting multiple future actions at once, but it also hard-codes a reactivity tradeoff. ACT introduced a transformer-based generative action-chunk policy for fine-grained bimanual manipulation [13], and Diffusion Policy showed that diffusion models can represent multi-modal visuomotor action distributions effectively [10]. RDT-1B scales diffusion-based action generation to bimanual foundation-model settings [32], while DexVLA represents a complementary VLA route that plugs a diffusion expert into a vision-language backbone [31]. These methods motivate the use of expressive action chunks, yet a fixed execution horizon can be brittle: short chunks can interrupt smooth progress, while long chunks can delay correction after contact changes, pose shifts, or miscalibrated predictions.

Adaptive execution horizons and test-time selection. Recent work has begun to relax fixed chunk execution. BID performs guided test-time sampling to improve action chunking by coupling forward and backward coherence [11]. Mixture-of-Horizons, VLA Knows Its Limits, Adaptive Action Chunking, HiPolicy, and related self-guidance methods explore learned or inference-time horizon choices using cross-horizon agreement, attention limits, entropy, hierarchical frequencies, or generated self-guidance [12, 33–36]. AHS differs in the evidence it uses and in where adaptation occurs: it derives intra-chunk evidence from the action expert’s generated trajectory, inter-chunk evidence from continuity between consecutive executions, and then maintains an online Beta posterior over candidate horizons. The QHA posterior used as an AHS prior provides an amortized initial preference, while AHS remains episode-local and can react to rollout-specific evidence.

E Limitations and Broader Impact

Limitations. AHS assumes that the frozen action expert exposes an action chunk and a velocity or denoising trace from which horizon-aware evidence can be computed. It also evaluates a discrete candidate set $\mathcal{K}$, so its resolution is limited by the chosen candidate grid. QHA amortizes a dense per-step prior, while AHS+QHA still combines it with interpolated sparse AHS online evidence. Our evaluation covers multiple backbones on RoboTwin2.0 and RoboCasa GR1 Tabletop and a real-world study on four bimanual household tasks, but not all robot embodiments, safety-critical tasks, or long-horizon mobile manipulation settings.

Broader impact. More reliable horizon adaptation may improve the robustness and efficiency of robot manipulation systems by reducing unnecessary replanning during stable motion while preserving reactivity near contact. The main risk is that an incorrectly selected horizon can still over-commit a robot to unsafe or undesired motion, especially outside the tested task distribution. Real-world use should therefore retain standard robot safeguards such as workspace limits, speed limits, emergency stops, human supervision during evaluation, and task-specific validation before deployment.

697 **F Existing Assets**

698 The experiments use existing benchmarks, model families, and codebases only for research evaluation.
699 RoboTwin2.0 and RoboCasa GR1 Tabletop are credited through their original papers [18, 19]. The
700 base policies and related codebases are credited through the cited work for π_0 , $\pi_{0.5}$, X-VLA, Fast-
701 WAM, Isaac-GR00T, and StarVLA [1–7]. We do not repackage these assets in the current draft. Any
702 released artifacts derived from this work will include the exact licenses, terms of use, and attribution
703 required by the corresponding upstream projects.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes].

Justification: The abstract and introduction state the AHS/QHA framework, assumptions, evaluation scope, and main empirical findings, which are supported by Sec. 4.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes].

Justification: The appendix discusses dependence on action-generation traces, discrete candidate horizons, benchmark coverage, and limited real-world coverage across four bimanual household tasks.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A].

Justification: The paper does not present theorem or proof claims; the mathematical material defines evidence terms and the online selection procedure.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes].

Justification: Sec. 4 and Appendix Sec. A specify benchmarks, tasks, base policies, rollout counts, candidate sets, and AHS/QHA configurations.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No].

Justification: Code and processed logs are not publicly released with the current submission; method details and evaluation protocols are provided in the paper and appendix.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes].

Justification: The experiments and Appendix Sec. A specify task suites, rollout protocols, base-policy checkpoints, hyperparameters, and QHA training settings.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No].

Justification: The paper reports fixed-rollout success rates and normalized process scores; Fig. 4 includes real-world error bars, but the main simulation tables do not report confidence intervals or significance tests.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes].

Justification: Appendix Sec. A reports the GPU classes used for simulation and the per-replan computational overhead of AHS.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes].

Justification: The authors are not aware of any deviation from the NeurIPS Code of Ethics; teleoperation is used only to collect robot demonstration trajectories and no privacy-sensitive data is collected.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes].

Justification: The appendix discusses potential robustness benefits for robot manipulation and risks from unsafe deployment when horizon choices are incorrect.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A].

Justification: The paper does not release a high-risk pretrained model, scraped dataset, or generative model artifact.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No].

Justification: The paper credits the benchmarks, models, and codebases through citations and appendix discussion, but does not enumerate every asset's exact license string or terms of use.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.

- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A].

Justification: The current submission does not release a new dataset, model checkpoint, or standalone asset package.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A].

Justification: The paper does not involve crowdsourcing or human-subject experiments; teleoperation is used only to collect robot demonstration trajectories.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A].

1016 Justification: No crowdsourcing or human-subject measurements are involved, so IRB
1017 approval or equivalent review is not applicable.

1018 Guidelines:

- 1019 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
1020 with human subjects.
- 1021 • Depending on the country in which research is conducted, IRB approval (or equivalent)
1022 may be required for any human subjects research. If you obtained IRB approval, you
1023 should clearly state this in the paper.
- 1024 • We recognize that the procedures for this may vary significantly between institutions
1025 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
1026 guidelines for their institution.
- 1027 • For initial submissions, do not include any information that would break anonymity (if
1028 applicable), such as the institution conducting the review.

1029 16. Declaration of LLM usage

1030 Question: Does the paper describe the usage of LLMs if it is an important, original, or
1031 non-standard component of the core methods in this research? Note that if the LLM is used
1032 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
1033 scientific rigor, or originality of the research, declaration is not required.

1034 Answer: [N/A].

1035 Justification: The core method does not use an LLM as an original or non-standard research
1036 component; VLA/VLM backbones are studied as robot policies.

1037 Guidelines:

- 1038 • The answer [N/A] means that the core method development in this research does not
1039 involve LLMs as any important, original, or non-standard components.
- 1040 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
1041 be described.